PM4Furniture: A Scriptable Parametric Modeling Interface for Conceptual Furniture Design Using PM4VR

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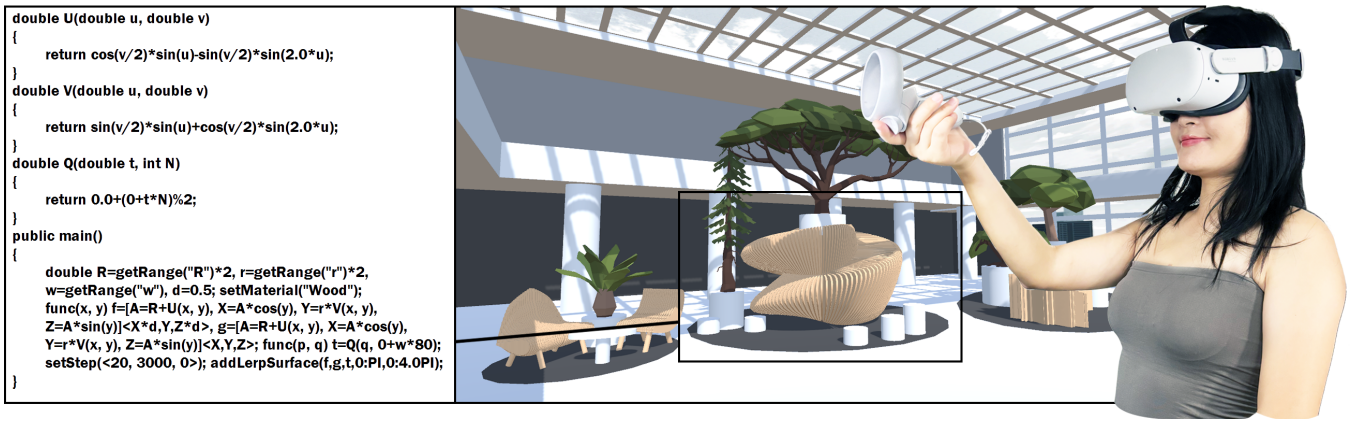


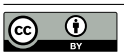
Figure 1: This teaser illustrates the application scenario of PM4Furniture, which allows users to input Java[b] script for defining and generating parametric furniture models (depicted in the left subfigure). Upon executing the script within the PM4Furniture interface, users can instantly visualize and adjust the parameters of resulting furniture's 3D model (highlighted in the middle subfigure) in a fully immersive VR setting, fostering a intuitive design experience (shown in the right subfigure).

Abstract

In the field of furniture design, Virtual Reality (VR) has shown potential in enabling immersive prototyping and visualization. However, current VR design tools are often limited by a lack of parametrization, making it challenging for designers to iterate complex furniture forms quickly. This paper presents PM4Furniture, a scriptable parametric modeling interface tailored for VR-based conceptual furniture design. The proposed system leverages a scripting interface of PM4VR framework with a low learning curve that allows designers to adjust the parameters of 3D furniture models in real time. By integrating a VR environment, PM4Furniture enhances user's interaction and intuitive adjustments of design parameters with immediate visual feedback. We evaluate this novel interface through a preliminary user study with designers interacting with furniture design in an immersive VR environment, revealing PM4Furniture's design efficiency and creativity in VR furniture prototyping.

CCS Concepts

• **Computer Graphics**; • **Virtual Reality**; • **Interactive System**;



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Keywords

Parametric modeling, virtual reality, furniture design, immersive design, scriptable interface

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1 Introduction

Conceptual design is the process of creating initial ideas and experimental forms for products, focusing on innovation, creativity, and exploration of new design possibilities. Conceptual furniture design, prioritizing furniture's artistic expression, is a phase where designers experiment with furniture's unique forms, shapes, materials, structures, and styles to push boundaries. Conceptual furniture design demands not only a high degree of flexibility but also an environment that encourages rapid experimentation and creative exploration. Traditional CAD software, though powerful [Dan 2017], often constrains designers to rigid workflows with static views, limited customization, and predefined component libraries. These limitations hinder the fluidity needed for early-stage ideation, where designers benefit from the freedom to modify shapes and envision different configurations in real-time [Ashraf 2020]. Furthermore, CAD software is typically optimized for precision and

detail [Scurtu et al. 2015], which can slow down the process of testing new creative ideas [Hamza Hamad 2020].

As an emerging technology, Virtual Reality (VR) presents a transformative alternative by enabling designers to step directly into the 3D world of their creations, fostering a deeper understanding of spatial relationships, ergonomics, and aesthetic appeal that 2D interfaces struggle to provide. [Jimeno-Morenilla et al. 2016; Roberts et al. 2020] VR's immersive qualities allow users to evaluate 3D conceptual designs at a real scale, make in-the-moment modifications, and experience a sense of presence when considering how a conceptual design looks and feels in a real space [Bernardo and Duarte 2022; Hamurcu et al. 2020]. To address these needs, this paper proposes PM4Furniture, a scriptable parametric modeling tool for conceptual furniture design that not only supports traditional parametric adjustments but also integrates these adjustments into an immersive hands-on design experience in VR. With PM4Furniture, designers can explore variations by tweaking design parameters while viewing changes within a virtual environment.

Fig. 1 illustrates the application scenario of PM4Furniture. The workflow begins with the Java[b] code as input to be executed within the PM4Furniture platform, the resulting parametric model is generated so as to allow the user to adjust parameters interactively. Then, users can manipulate the furniture model in a virtual setting, in this instance representing a lobby space, providing a realistic spatial context where designers can assess the scale, form, and aesthetic coherence of the conceptual furniture elements in real time.



Figure 2: Examples of Parametric Furniture Design. These parametric furniture pieces located in in St. Petersburg are designed by Denis Homyakov [Homyakov 2020] and produced by the P.metric Studio [P.metric 2017].

2 Related Work

Parametric furniture design represents a unique and functional design approach that can enhance interior spaces with a modern interpretation of furniture that embraces technology and innovation. Utilizing parametric design principles, this novel furniture design approach leverages algorithms and computational design processes to create aesthetically pleasing furniture pieces that are adaptable to various environments and functional needs. As illustrated in Fig. 2, several examples of parametric furniture designed by Denis Khomyakov [Homyakov 2020] and produced by P.metric Studio [P.metric 2017] can be found in the recreation area along the embankment of the Moika River in St. Petersburg. These furniture exemplify parametric design's ability to generate complex shapes that challenge traditional aesthetics while providing functionality.

While reshaping the field of furniture design, Virtual Reality (VR) is providing a powerful tool for designers and consumers to visualize, interact with, and refine furniture pieces in a virtual space before production. Fleury et al. [Fleury et al. 2022] explore VR's role in improving design accuracy, enhancing customer engagement, and streamlining iterative feedback. Prabhakaran et al. [Prabhakaran et al. 2021] conduct an empirical study examining the effectiveness of interactive virtual reality (VR) in facilitating communication related to the design of furniture, fixtures, and equipment. Similarly, Xin et al. [Xin 2024] investigate VR-based product demonstrations specifically for furniture items, showcasing how such virtual experiences can effectively convey the features and aesthetics of products to potential buyers. Meanwhile, Saixiong et al. [Saixiong et al. 2024] focus on VR applications for Ming-style furniture designs.

Recent research driven by the PM4VR interface proposed by Li et al. [Li 2022] demonstrates a growing trend in applying VR platforms across diverse design disciplines, including architectural design, industrial design, and artistic design. PM4VR serves as a foundational framework for a variety of creative applications, facilitating conceptual design tasks that permeate everyday life. For instance, PM4VR has been extended for car design [Li 2024b], allowing designers to create and visualize automotive concepts in immersive environments. Similarly, PM4VR has been extended to fashion design [Li 2024c] and bag design [Li 2024a], enabling designers to experiment with garments and accessories in a virtual setting. PM4VR has also been extended to pottery design [Li 2024g], wire art design [Li 2024k], wall art design [Li 2024j] and decorative plants design [Li 2024d,e,h], illustrating its adaptability in different artistic domains. Furthermore, PM4VR has extended its applications to urban layout design [Li 2024i] and music visualizer design [Li 2024f], showcasing its potential to transform how designers approach their work by offering a more interactive and engaging design experience. Inspired by these existing works, we propose PM4Furniture: a scriptable parametric modeling interface tailored for conceptual furniture design in VR, targeted at enhancing creative exploration and rapid iteration by bringing users a new level of control to seamlessly transition from an easy-to-write script input to an interactive, immersive, and manipulable VR prototype.

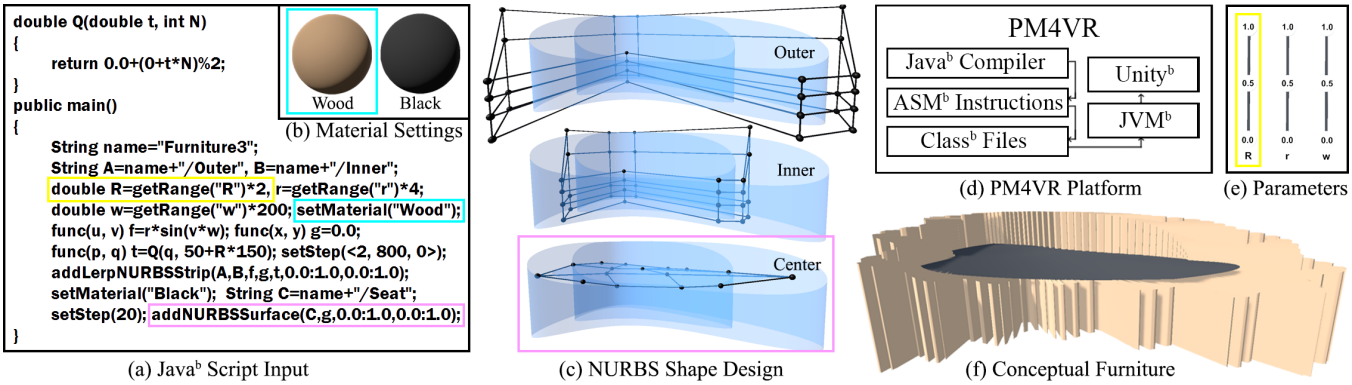


Figure 3: Overview of our approach.

3 Overview

Fig. 3 shows the overall streamline of PM4Furniture for conceptual furniture design in VR is outlined. As shown in Fig. 3(a), user designers can script their furniture design using Java^b language so as to easily outline and define the parametric functions of the furniture design. Moving on to Fig. 3(b), PM4Furniture is seamlessly integrated into Unity3D Editor, so that users can easily edit materials in Unity3D Editor. Step in Fig. 3(c) involves the PM4Furniture's component of NURBS Editor [Li 2024b,c,g] to construct and manipulate the furniture's shape with precision and customization. Then, the PM4VR platform [Li 2022] shown in Fig. 3(d) serves as the core VR environment for PM4Furniture where all components join together, allowing designers to interact with and visualize the evolving furniture model in an immersive VR setting. After execution, the VR parameter panel shown in Fig. 3(e) is synthesized for adjustable variables that control the furniture's geometric features. Finally, conceptual furniture designed with PM4Furniture is shown in Fig. 3(f), presenting an end output ready for further production.

4 Technical Approach

To address the specific needs of conceptual furniture design, Parametric Linear Interpolation (Lerp) algorithm, a crucial feature absent in PM4VR and its extensions, was implemented in PM4Furniture. This section introduces the mathematics behind this algorithm.

In conceptual furniture design, there is a requirement for high-frequency alternating interpolation between two surfaces. Mathematically, given two surfaces geometry which are defined with parametric function $\mathbf{f}(u,v) \in \mathbb{R}^3$, $\mathbf{g}(u,v) \in \mathbb{R}^3$ and a Lerp parametric function $t(u,v) \in [0,1]$ through two parameters u and v where $u \in [u_0,u_1]$, $v \in [v_0,v_1]$, $u_0 < u_1 \in \mathbb{R}$, and $v_0 < v_1 \in \mathbb{R}$. Then, a Lerp surface $\mathbf{k}(u,v) = (1-t(u,v))\mathbf{f}(u,v) + t(u,v)\mathbf{g}(u,v)$. In PM4Furniture, we extend the PM4VR with a new Java^b method of: **addLerpSurface(** $\mathbf{f}(u,v)$, $\mathbf{g}(u,v)$,$t(u,v)$,$u_0 : u_1$, $v_0 : v_1$). Fig.4 shows an example of PM4Furniture by adding Lerp surface and assigning Lerp parametric function as a rectangular wave function $Q(t,N) = \lfloor t \cdot N \rfloor \mod 2$, where N is the number of slices.

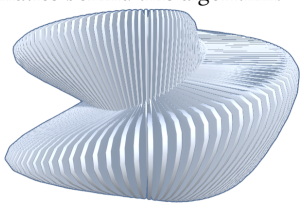


Figure 4: Lerp Surface.

As shown in Fig. 5, PM4Furniture extends the PM4VR with a functionality of interpolation between two NURBS surfaces by providing with a new Java^b method called: **addLerpNURBSSurface(** $\langle A\rangle$, $\langle B\rangle$, $f(u,v)$, $g(u,v)$, $t(u,v)$, $u_0 : u_1, v_0 : v_1$), where "A.NURBS" and "B.NURBS" are two files storing the control points information for two NURBS surfaces of A and B whose parametric equations are $\mathbf{s}^{A|B}(u,v)$ which are calculated as :

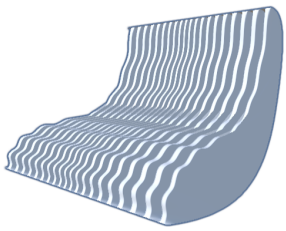


Figure 5: Lerp NURBS.

$$\mathbf{s}^{A|B}(u,v) = \frac{\sum_{i=0}^{m}\sum_{j=0}^{n} w_{i,j}^{A|B} b_p^i(u) b_q^j(v) \mathbf{p}_{i,j}^{A|B}}{\sum_{i=0}^{m}\sum_{j=0}^{n} w_{i,j}^{A|B} b_p^i(u) b_q^j(v)}, \tag{1}$$

where where $b_p^i(u)$ is the pth order B-spline basis function of the ith control point, $\mathbf{p}_{i,j}^{A|B}$ and $w_{i,j}^{A|B}$ are the control points positions and weights of NURBS surface A and B. Then, the Lerp NURBS surface is calculated as: $\mathbf{k}(u,v) = (1-t(u,v))\mathbf{s}^A \diamond f(u,v) + t(u,v)\mathbf{s}^B \diamond g(u,v)$, where surface normal displacement operator $\diamond$ is defined as:

$$\mathbf{s} \diamond f(u,v) = \mathbf{s}(u,v) + \frac{\mathbf{s}_u(u,v) \times \mathbf{s}_v(u,v)}{\left\|\mathbf{s}_u(u,v) \times \mathbf{s}_v(u,v)\right\|} f(u,v) \tag{2}$$

As shown in Fig. 6, PM4Furniture extends the PM4VR with a functionality of interpolation between two symmetric NURBS surfaces through a new Java^b method of: **addLerpNURBSSymmetry(** $\langle A\rangle$, $\langle B\rangle$, $f(u,v)$, $g(u,v)$, $t(u,v)$, $\mathbf{q}$, $u_0 : u_1$, $v_0 : v_1$), where A and B are those two NURBS surfaces whose parametric equations are $\mathbf{s}^{A|B}(u,v)$. Then, the Lerp NURBS symmetry is calculated as:

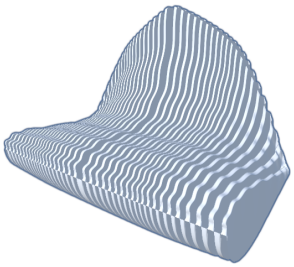


Figure 6:Lerp Symmetry

$$\mathbf{k}_{\mathbf{q}}(u,v) = \mathbf{q} \circ \Big((1-t(u,v))\mathbf{s}^A \diamond f(u,v) + t(u,v)\mathbf{s}^B \diamond g(u,v) \Big) \tag{3}$$

where the vector scaling operator $\circ$ means: $\langle x_1,y_1,z_1\rangle \circ \langle x_2,y_2,z_2\rangle = \langle x_1x_2, y_1y_2, z_1z_2\rangle$. Here, $\mathbf{q}$ specifies the symmetric plane of the NURBS surface. When $\mathbf{q} = \langle -1,1,1\rangle$, $\mathbf{q} = \langle 1,-1,1\rangle$ or $\mathbf{q} = \langle 1,1,-1\rangle$, the Lerp surface will be flipped along x, y or z-axis.

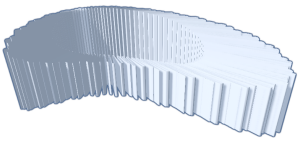


Figure 7: Lerp Strip.

As shown in Fig. 7, PM4Furniture extends the PM4VR with a functionality of interpolation between two NURBS strips by providing with a new Java[♭] method called: **addLerpNURBSStrip**($\langle A \rangle$, $\langle B \rangle$, $f(u,v)$, $g(u,v)$, $t(u,v)$, $u_0 : u_1$, $v_0 : v_1$), where "A.NURBS" and "B.NURBS" are two files storing the control points information for two NURBS strips of A and B whose parametric equations $\mathbf{s}^{A|B}(u,v)$ are calculated with following equation:

$$\mathbf{s}^{A|B}(u,v) = \frac{\sum_{i=0}^{m}\sum_{j=0}^{n+q} w^{A|B}_{i,j \pmod n} b^i_p(u) b^j_q(v) \mathbf{p}^{A|B}_{i,j \pmod n}}{\sum_{i=0}^{m}\sum_{j=0}^{n+q} w^{A|B}_{i,j \pmod n} b^i_p(u) b^j_q(v)} \quad (4)$$

The, the Lerp NURBS strip can be calculated with this equation: $\mathbf{k}(u,v) = (1 - t(u,v))\mathbf{s}^A \diamond f(u,v) + t(u,v)\mathbf{s}^B \diamond g(u,v)$.

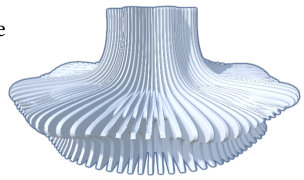


Figure 8: Lerp Bottle.

As shown in Fig. 7, PM4Furniture extends the PM4VR with a functionality of interpolation between two NURBS bottles by providing with a new Java[♭] method called: **addLerpNURBSBottle**($\langle A \rangle$, $\langle B \rangle$, $f(u,v)$, $g(u,v)$, $t(u,v)$, $u_0 : u_1$, $v_0 : v_1$), where "A.NURBS" and "B.NURBS" are two files storing the control points information for two NURBS bottles of A and B whose parametric equations $\mathbf{s}^{A|B}(u,v)$ are calculated as:

$$\mathbf{s}^{A|B}(u,v) = \begin{bmatrix} r^{A|B}(t)\cos\theta \\ \mathbf{q}^{A|B}(t)\cdot\hat{\mathbf{y}} \\ r^{A|B}(t)\sin\theta \end{bmatrix}, \quad \mathbf{q}^{A|B}(t) = \frac{\sum_{i=1}^{n} w_i^{A|B} b_p^i(t) \mathbf{p}_i^{A|B}}{\sum_{i=1}^{n} w_i^{A|B} b_p^i(t)},$$

where $r^{A|B}(t) = ||\mathbf{q}^{A|B}(t) - \hat{\mathbf{y}}(\mathbf{q}^{A|B}(t)\cdot\hat{\mathbf{y}})||$, $t = (u-u_0)/(u_1-u_0)$, $\theta = 2\pi(v-v_0)/(v_1-v_0)$. Lerp NURBS bottle is calculated as: $\mathbf{k}(u,v) = (1-t(u,v))\mathbf{s}^A \diamond f(u,v) + t(u,v)\mathbf{s}^B \diamond g(u,v)$.

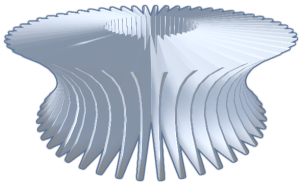


Figure 9: Lerp Pipe.

In PM4VR, a pipe geometry is constructed with curve's position $\mathbf{f}(\lambda)$ and curve's radius $g(\lambda)$. As shown in Fig. 9, PM4Furniture extends the PM4VR with a functionality of interpolation between two pipes by providing with a new Java[♭] method: **addLerpPipe** ($\mathbf{f}_1(\lambda)$, $\lambda_{1,0} : \lambda_{1,1}$, $g_1(\lambda)$, $\lambda'_{1,0} : \lambda'_{1,1}$, $\mathbf{f}_2(\lambda)$, $\lambda_{2,0} : \lambda_{2,1}$, $g_2(\lambda)$, $\lambda'_{2,0} : \lambda'_{2,1}$, $t(u,v)$, $u_0 : u_1$, $v_0 : v_1$). Then, the pipe geometry is calculated as:

$$\mathbf{p}_{1|2}(u,v) = \mathbf{f}_{1|2}(\lambda_{1|2,u}) + g_{1|2}(\lambda'_{1|2,u}) \mathfrak{U}^{\hat{\mathbf{f}'}_{1|2}(\lambda_{1|2,u})}_{\theta_v} \left(\hat{\mathbf{f}'}_{1|2}(\lambda_{1|2,u})_\perp\right),$$

where there are: $\hat{\mathbf{n}} = \frac{\mathbf{n}}{||\mathbf{n}||}$, $\mathbf{f}'(\tau) = \frac{\mathrm{d}\mathbf{f}(\tau)}{\mathrm{d}\tau}$, angle-axis rotation function $\mathbf{v}' = \mathfrak{U}^{\mathbf{n}}_{\alpha}(\mathbf{v})$ returns to a vector $\mathbf{v}'$ from rotating $\mathbf{v}$ along axis $\mathbf{n}$ with angle α, perpendicular notation $\perp$ means $\mathbf{v}\cdot\mathbf{v}_\perp = 0$, and:

$$\lambda_{1|2,u} = \lambda_{1|2,0}\left(\frac{u1-u}{u_1-u_0}\right) + \lambda_{1|2,1}\left(\frac{u-u_0}{u_1-u_0}\right), \theta_v = 2\pi\left(\frac{v-v_0}{v_1-v_0}\right).$$

Then, Lerp pipe is: $\mathbf{k}(u,v) = (1-t(u,v))\mathbf{p}_1(u,v) + t(u,v)\mathbf{p}_2(u,v)$.

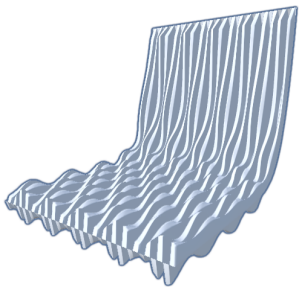


Figure 10: Lerp Crust.

In MP4VR, a crust surface is constructed by extruding any surface $\mathbf{f}_(u,v)$ along its normal direction at a specific distance $g(u,v)$ and opposite direction at a specific distance $q(u,v)$ as two sides of a crust surface. As shown in Fig. 10, PM4Furniture extends the PM4VR with a functionality of interpolation between a crust surface's two sides by a new Java[♭] method: **addLerpCrust**($\mathbf{f}(u,v)$, $u_0 : u_1$, $v_0 : v_1$, $g(u,v)$, $u'_0 : u'_1$, $v'_0 : v'_1$, $q(u,v)$, $u''_0 : u''_1$, $v''_0 : v''_1$, $t(u,v)$, $u'''_0 : u'''_1$, $v'''_0 : v'''_1$). Then, Lerp crust is: $\mathbf{k}(u,v) = (1-t(u,v))\mathbf{f} \diamond g(u,v) + t(u,v)\mathbf{f} \diamond (-q)(u,v)$.

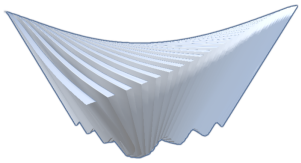


Figure 11: Lerp Coons.

In PM4VR, Coons surfaces are generated from four borders defined with four curve equations which are $\mathbf{p}_{U0}(u)$, $\mathbf{p}_{U1}(u)$, $\mathbf{p}_{0V}(v)$, and $\mathbf{p}_{0V}(v)$ by the linear interpolation between two opposite borders subtracted by the bilinear interpolation between four borders. As shown in Fig. 11, PM4Furniture extends the PM4VR with a functionality of interpolation between two Coons surfaces by a new Java[♭] method: **addLerpCoonsSurface** ($\mathbf{p}_{U0}(u)$, $u_0 : u_1$, $\mathbf{p}_{U1}(u)$, $u'_0 : u'_1$, $\mathbf{p}_{0V}(v)$, $v_0 : v_1$, $\mathbf{p}_{1V}(v)$, $v'_0 : v'_1$, $\mathbf{q}_{U0}(u)$, $u''_0 : u''_1$, $\mathbf{q}_{U1}(u)$, $u'''_0 : u'''_1$, $\mathbf{q}_{0V}(v)$, $v''_0 : v''_1$, $\mathbf{q}_{1V}(v)$, $v'''_0 : v'''_1$, $t(u,v)$, $u''''_0 : u''''_1$, $v''''_0 : v''''_1$). Then, let $\check{\mathbf{p}}(u,v)$ represents the Coons surface constructed by following equations:

$$\begin{aligned}\check{\mathbf{p}}(u,v) = &\begin{bmatrix}1-u & u\end{bmatrix}\begin{bmatrix}\mathbf{p}_{0V}(v)\\ \mathbf{p}_{1V}(v)\end{bmatrix} + \begin{bmatrix}\mathbf{p}_{U0}(u) & \mathbf{p}_{U1}(u)\end{bmatrix}\begin{bmatrix}1-v\\ v\end{bmatrix} \\ &- \begin{bmatrix}1-u & u\end{bmatrix}\begin{bmatrix}\mathbf{p}_{00} & \mathbf{p}_{01}\\ \mathbf{p}_{10} & \mathbf{p}_{11}\end{bmatrix}\begin{bmatrix}1-v\\ v\end{bmatrix}\end{aligned} \quad (5)$$

Lerp Coons Surface is: $\mathbf{k}(u,v) = (1-t(u,v))\check{\mathbf{p}}(u,v) + t(u,v)\check{\mathbf{q}}(u,v)$.

5 Experiment Results

This section demonstrate the experiments results that underscore PM4Furniture's effectiveness in generating various types of conceptual furniture design and its strengths in fostering creativity. Fig. 12-14 illustrate the designer's input, which consists of Java[♭] scripts, alongside the NURBS shape designs, and the outputs of corresponding conceptual furniture generated by PM4Furniture.

The first conceptual furniture's geometry is constructed with a Lerp surface interpolated from two Klein surfaces $\mathbf{f}(x,y)$ and $\mathbf{g}(x,y)$, where $\mathbf{f}(x,y)$ is the inner surface with a smaller radius (d=0.5) whileg(x,y) is the outer surface with a larger radius (d=1.0). The second conceptual furniture's geometry is constructed with a Lerp NURBS surface interpolated from two NURBS surfaces which are "Top" and "Bottom" respectively. As shown in the top-left subfigure, these two NURBS surfaces are manually designed with the Unity NURBS Editor Plugin [Li 2024b]. These two NURBS surfaces are further displaced along the surface normal direction with the distance of sine function $r\sin(vw)$. The third conceptual furniture's geometry is constructed with a Lerp NURBS bottle interpolated from two NURBS bottles which are "Outer" and "Inner" respectively whose NURBS bottles are manually designed with the Unity

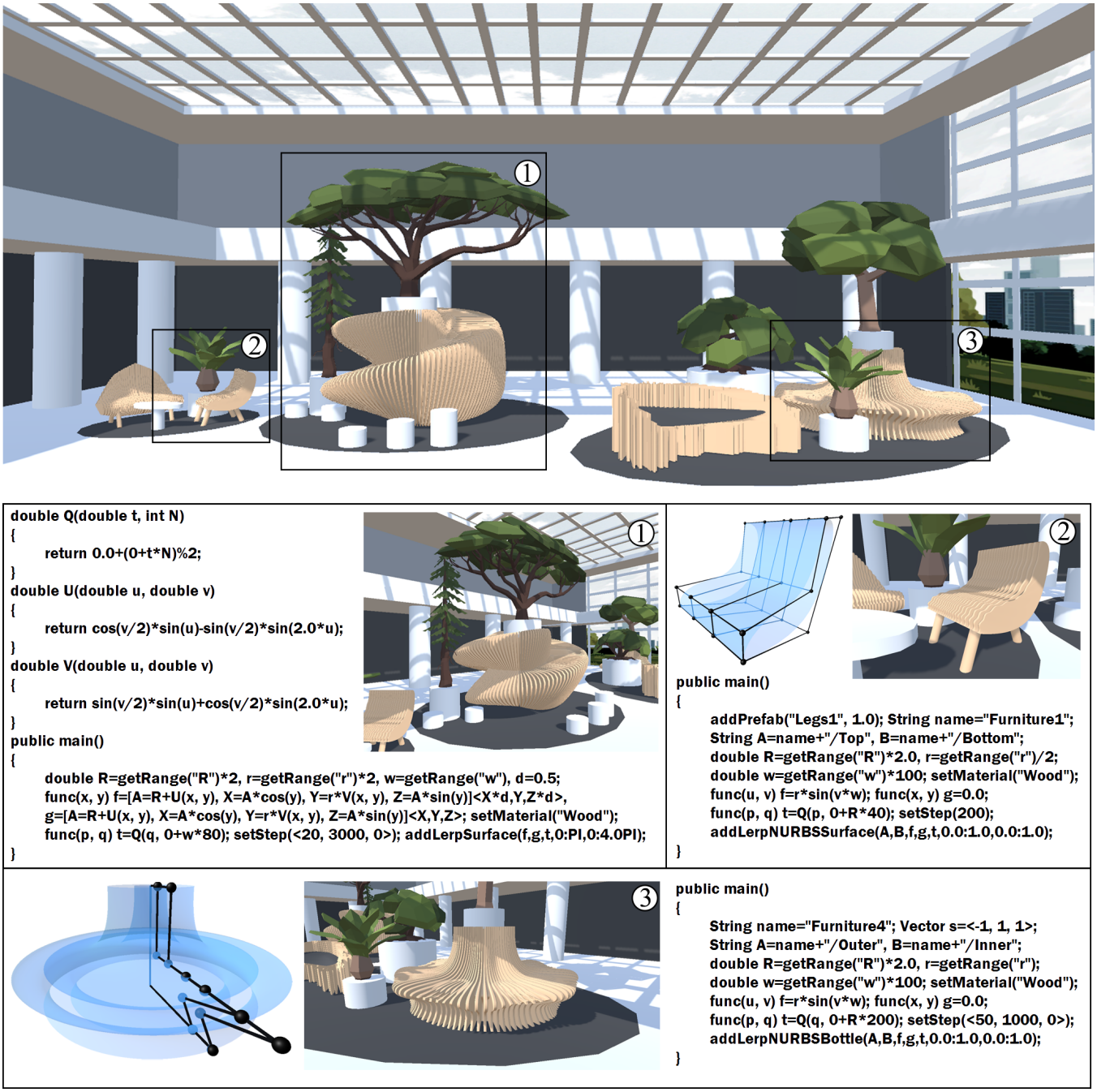


Figure 12: Experiment Results (Part 1). This figure shows 1st to 3rd conceptual furniture designs using PM4Furniture.

NURBS Editor Plugin [Li 2024g]. These two NURBS bottles are further displaced along the surface normal direction with the distance of sine function $r\sin(vw)$. As we can see from the result, this conceptual furniture can be functional as a circular chair in an open large recreation area or workspace such as a garden or lobby. Similar to the second conceptual furniture design, the fourth conceptual furniture's geometry is constructed with a Lerp NURBS symmetry whose NURBS surfaces are only designed for half of the chair, while the other half is automatically synthesized by flipping along the x-axis around the YZ-plane according to the scaling vector $\mathbf{q} = \langle -1, 1, 1\rangle$. Similar to the third conceptual furniture design, the fifth conceptual furniture's geometry is constructed with a Lerp NURBS strip interpolated from two NURBS strips which are manually designed with the Unity NURBS Editor Plugin [Li 2024c] and

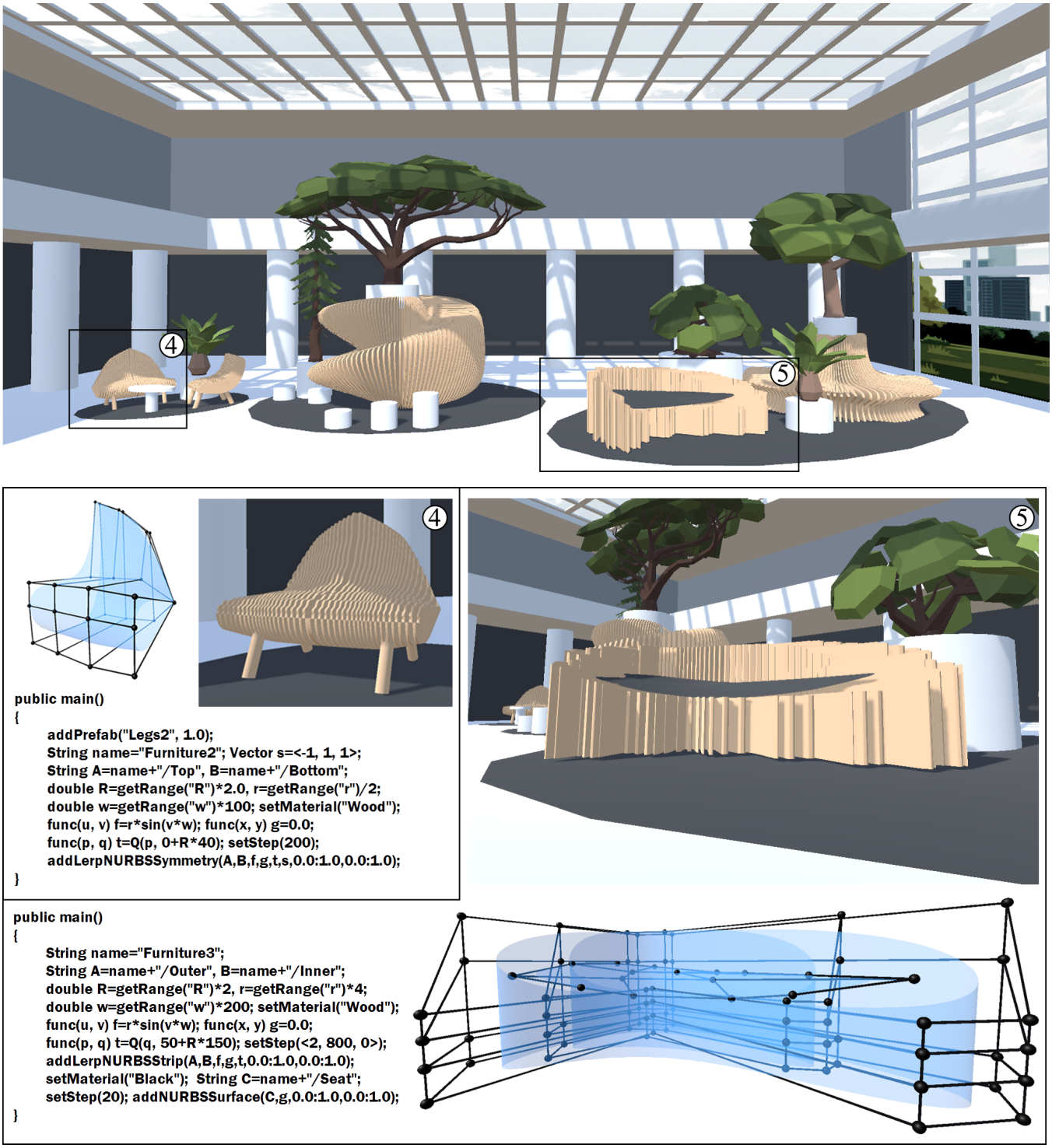


Figure 13: Experiment Results (Part 2). This figure shows 4th and 5th conceptual furniture designs using PM4Furniture.

displaced along the surface normal direction with the distance of sine function $r\sin(vw)$. The sixth conceptual furniture's geometry is constructed with a Lerp crust interpolated from two sides of a crust among which one's normal displacement function is $g(u, v)$ while another one's is $q(u, v)$. The seventh conceptual furniture's geometry is constructed with a Lerp pipe interpolated from two

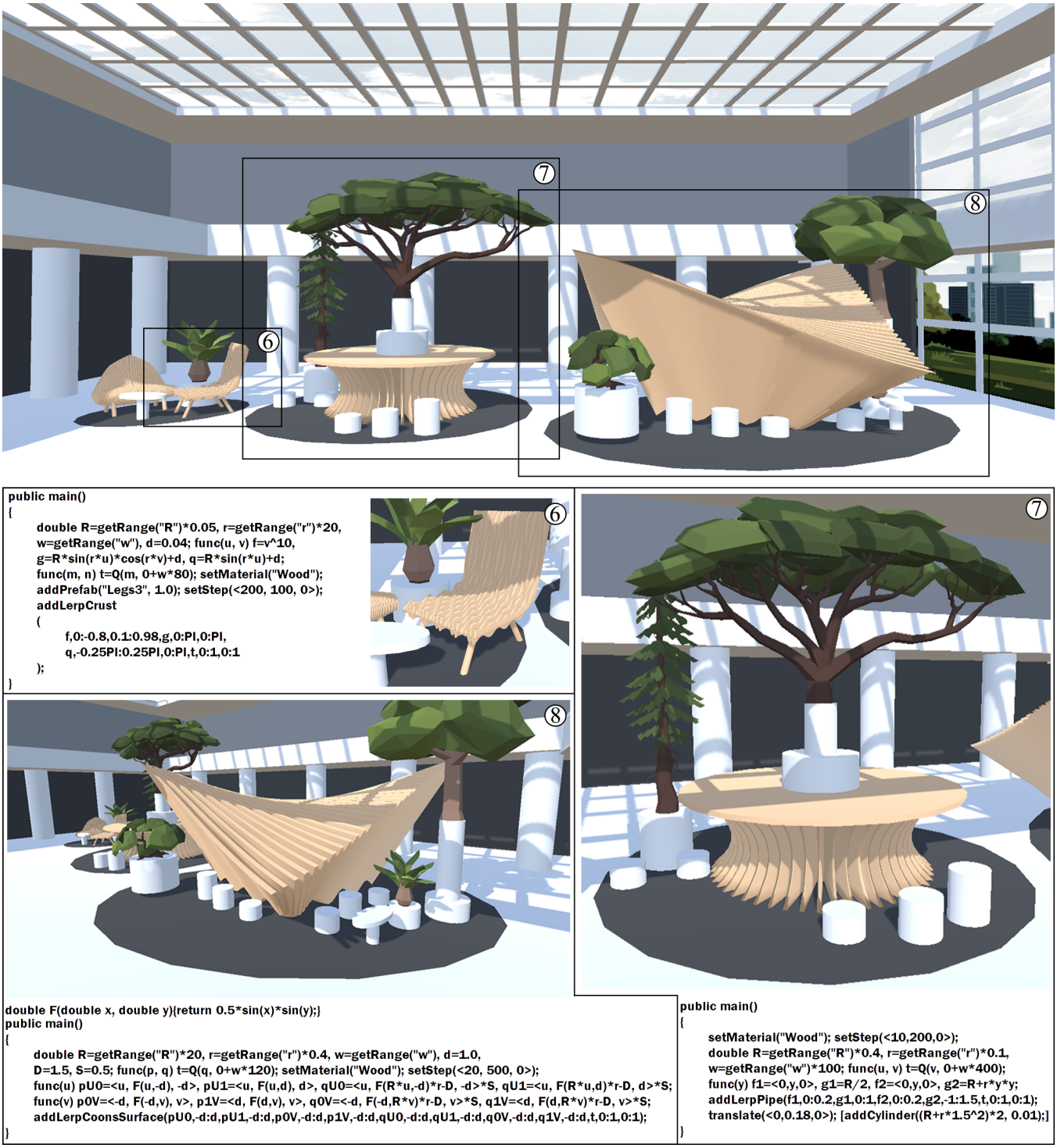


Figure 14: Experiment Results (Part 3). This figure shows 6th to 8th conceptual furniture designs using PM4Furniture.

pipes whose inner surface is a cylinder with a radius of $R/2$ and outer surface with a radius varying along the y-axis which is $R+ry^2$. The eighth conceptual furniture's geometry is constructed with a Lerp Coons surface interpolated from two Coons surfaces among which one is $F(x, y) = 0.5\sin(x)\sin(x)$ and another one is a transformation of the former one by translating it along y-axis with distance $D = -1.5$ and scaling it with $S = 0.5$.

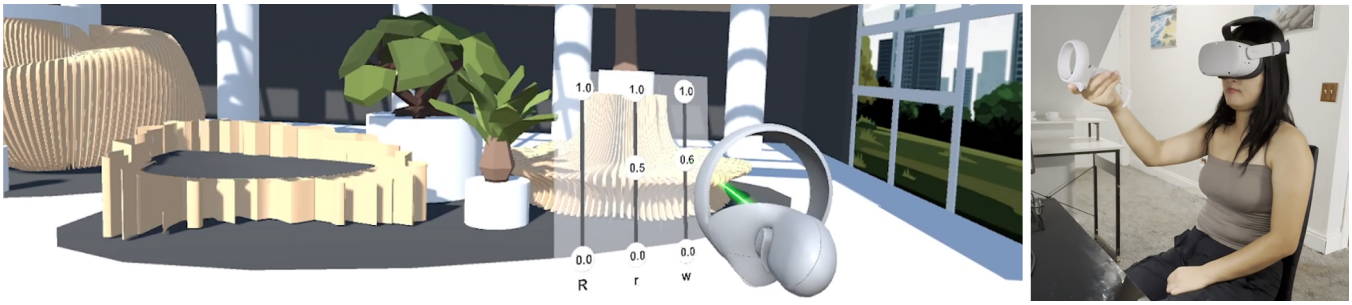


Figure 15: User Study. This figure shows a user using VR controllers to adjust furniture design parameters within PM4Furniture.

Fig. 15 shows a user wearing an Oculus Quest 2 VR headset and using VR controllers to adjust furniture design parameters within the PM4Furniture interface. In order to enhance the visual impact, coherence of the design, and immersion of the virtual scene, all conceptual furniture pieces are placed together within a unified virtual lobby environment which is decorated with indoor plants and other elements that contribute to a harmonious interior aesthetic. By adjusting the furniture design parameters within an immersive virtual space, the output integrated with visual feedback showcases how these designs might appear in real-world settings. Please find more details in this video recording: https://youtu.be/0eqLou3Owk0

6 Conclusion

In this paper, we propose PM4Furniture: a scriptable parametric modeling interface tailored for conceptual furniture design in VR, enabling users to seamlessly transition from script input to an interactive VR prototype. PM4Furniture's programming interface brings a new level of control, enabling designers to customize parameters through VR controllers, allowing intuitive interaction for adjusting conceptual furniture models dynamically. PM4Furniture's VR setup not only presents an impressive blueprint of interior design possibilities but also highlights the potential of this interface as a tool for envisioning creative, cohesive, and adaptable furniture designs for modern interiors. PM4Furniture offers a novel solution that enhances creative exploration and rapid iteration, bridges the gap between initial concept and practical design, and paves the way for a future where furniture is not just functional but also a reflection of personal style, mathematics, and creativity.

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